

## Lesson 02 Demo 03

### Validating Misconfigurations in Security Groups

**Objective:** To create and configure a security group and validate its compliance and configurations using AWS Trusted Advisor checks to ensure secure and optimized AWS resources

**Tools required:** AWS Management Console

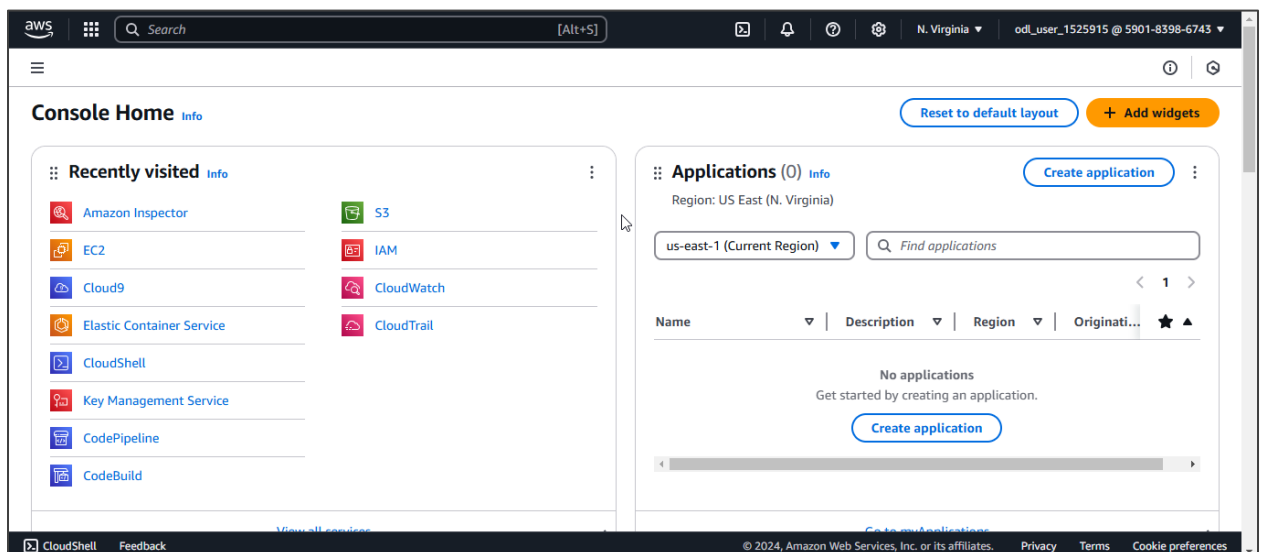
**Prerequisites:** None

Steps to be followed:

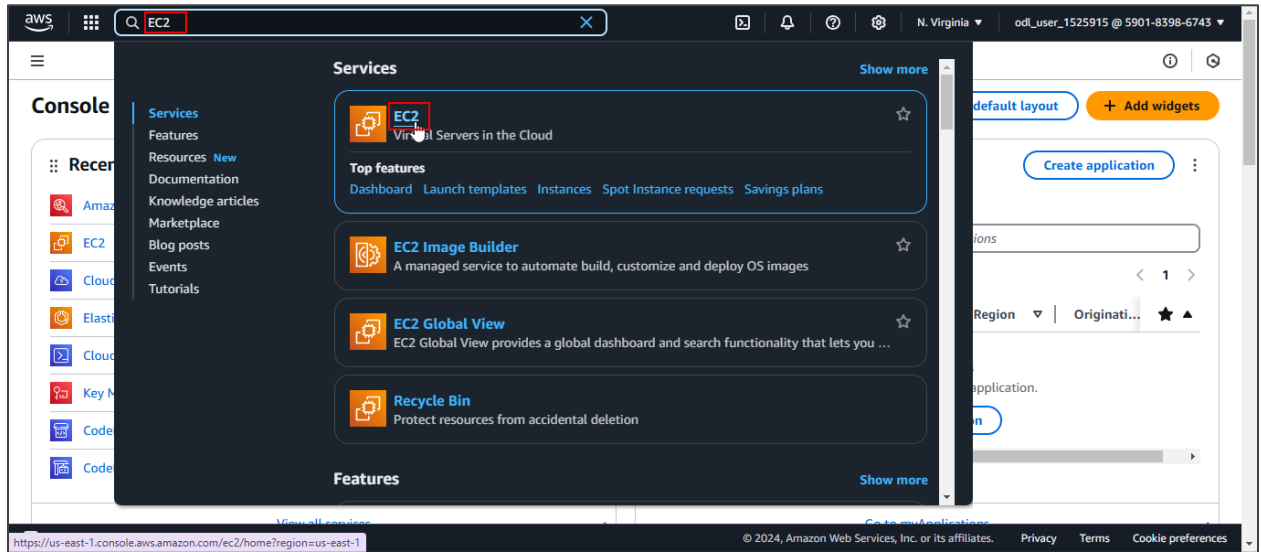
1. Create and configure a security group
2. Validate security configurations using AWS Trusted Advisor checks

#### Step 1: Create and configure a security group

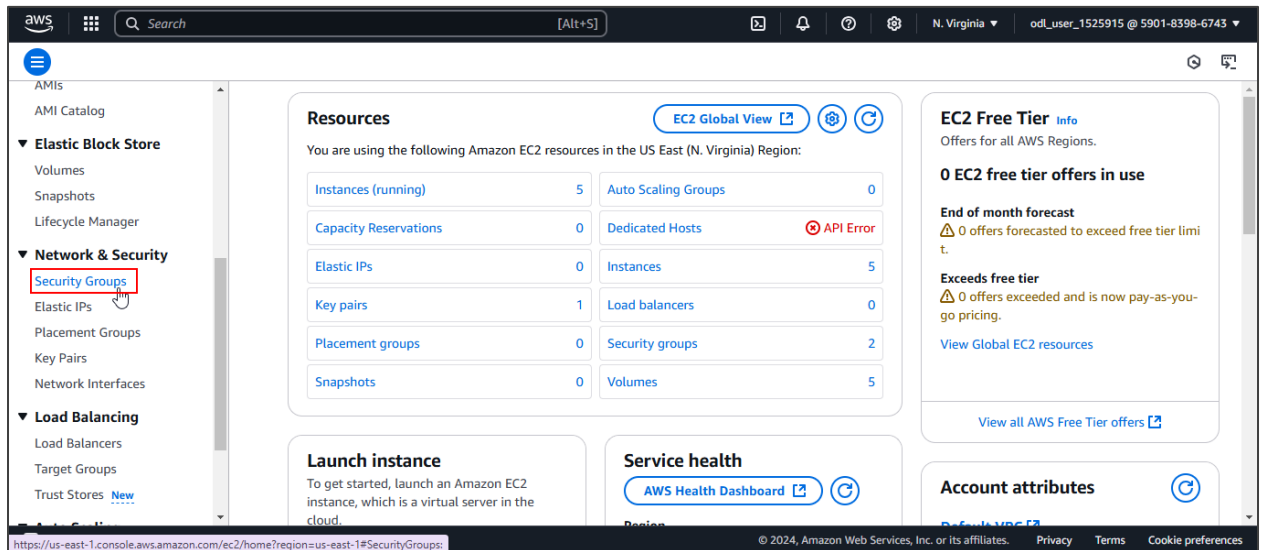
##### 1.1 Navigate to the AWS Console Home page



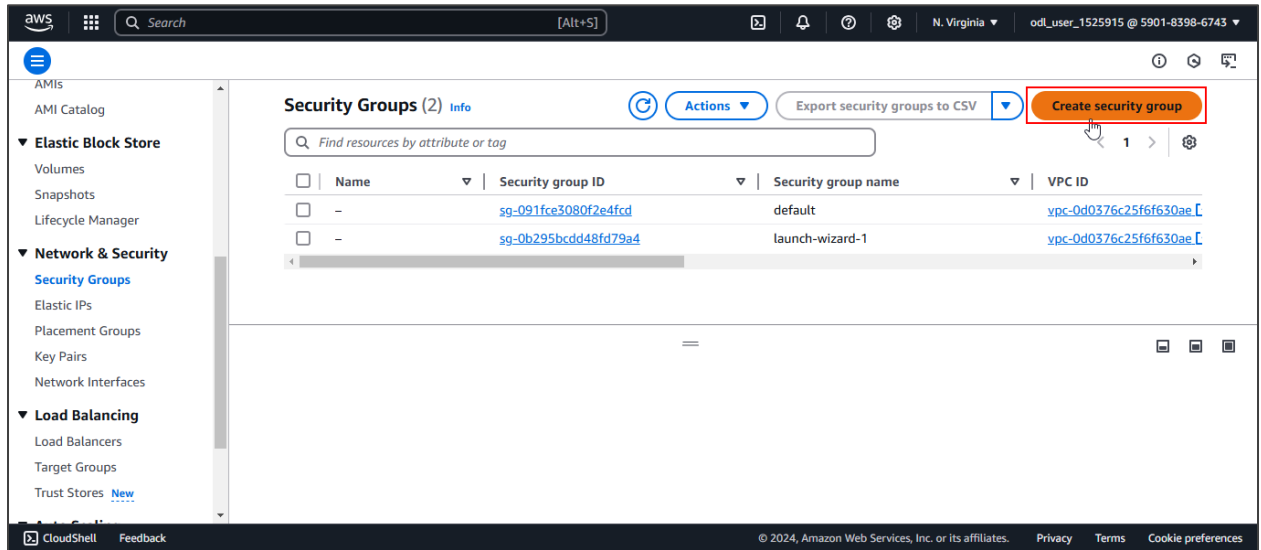
## 1.2 Search for and click on EC2



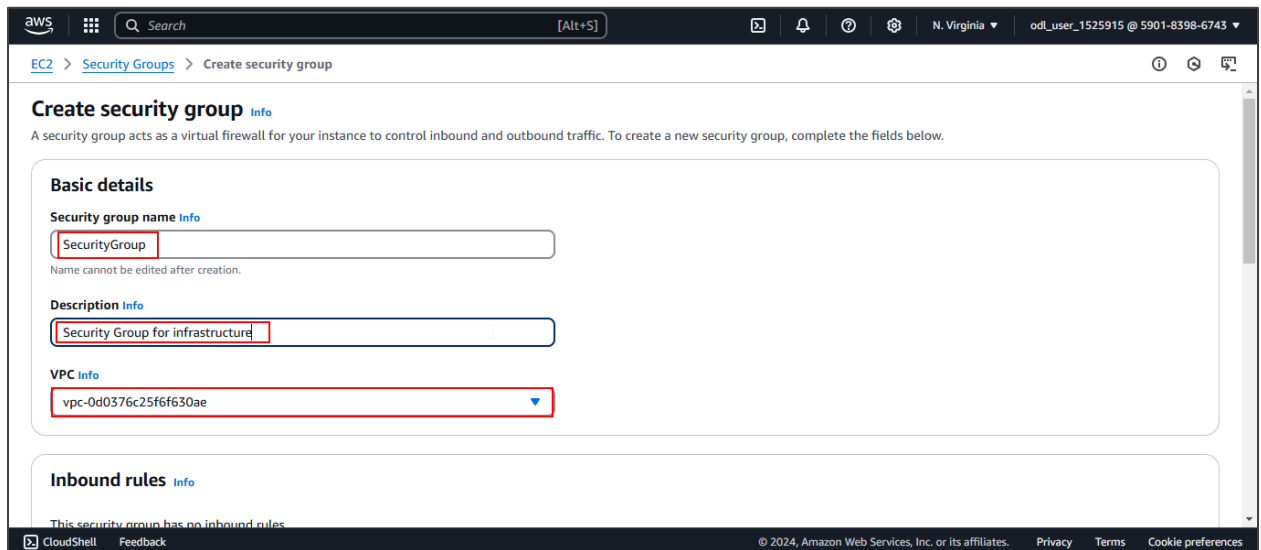
## 1.3 Click on Security Groups in the Network & Security section



## 1.4 Click on **Create security group**



## 1.5 Add **SecurityGroup** as the security group name and **Security Group for infrastructure** as the description. Then, select the desired VPC.



1.6 Configure outbound rules for **SSH (TCP/22 to 0.0.0.0/0)**, **RDP (TCP/3389 to 0.0.0.0/0)**, and **LDAP (TCP/389 to My IP:<YOUR\_IP>)**

The screenshot shows the 'Create security group' page in the AWS Management Console. The 'Outbound rules' section is active, displaying a table with three rules: SSH, RDP, and LDAP. Each rule is configured with its respective protocol, port, and destination. The SSH rule is set to TCP, port 22, and destination 0.0.0.0/0. The RDP rule is set to TCP, port 3389, and destination 0.0.0.0/0. The LDAP rule is set to TCP, port 389, and destination 'My IP' with a specific IP address 103.159.11.170/32. A red box highlights the rule configuration area. A warning message at the bottom states: 'Rules with destination of 0.0.0.0/0 or ::/0 allow your instances to send traffic to any IPv4 or IPv6 address. We recommend setting security group rules to be more restrictive and to only allow traffic to specific known IP addresses.'

| Type | Protocol | Port range | Destination       | Description - optional |
|------|----------|------------|-------------------|------------------------|
| SSH  | TCP      | 22         | 0.0.0.0/0         |                        |
| RDP  | TCP      | 3389       | 0.0.0.0/0         |                        |
| LDAP | TCP      | 389        | 103.159.11.170/32 |                        |

**Note:** Replace **<YOUR\_IP>** with your actual IP; in this case it is **103.159.11.170/32**

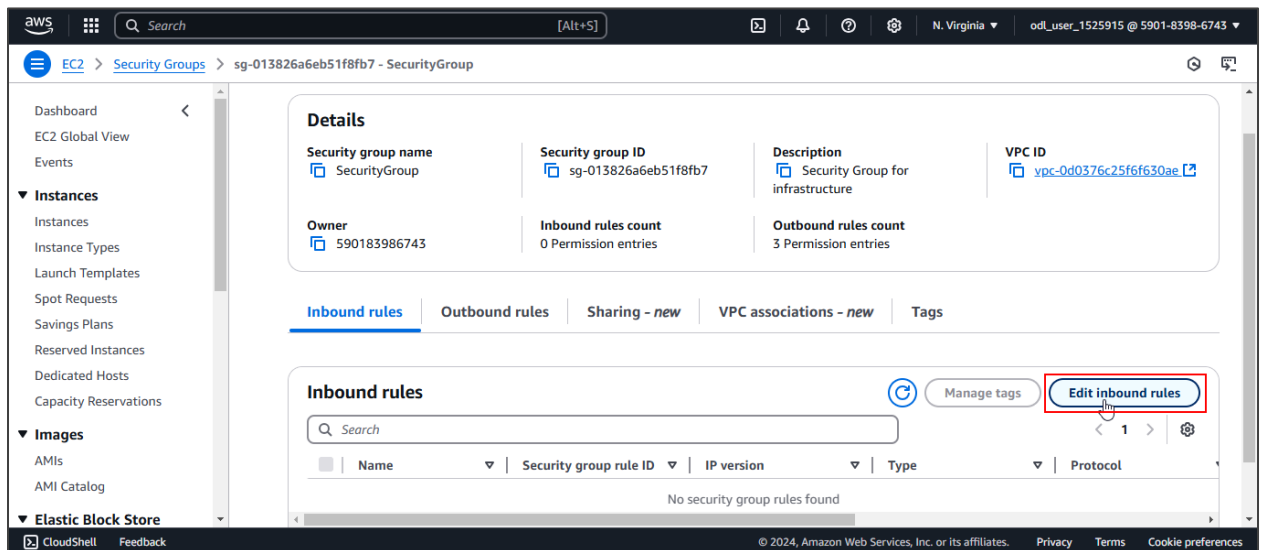
1.7 Click on **Create security group**

The screenshot shows the 'Create security group' page in the AWS Management Console. The 'Add rule' button is visible. A warning message at the bottom states: 'Rules with destination of 0.0.0.0/0 or ::/0 allow your instances to send traffic to any IPv4 or IPv6 address. We recommend setting security group rules to be more restrictive and to only allow traffic to specific known IP addresses.'

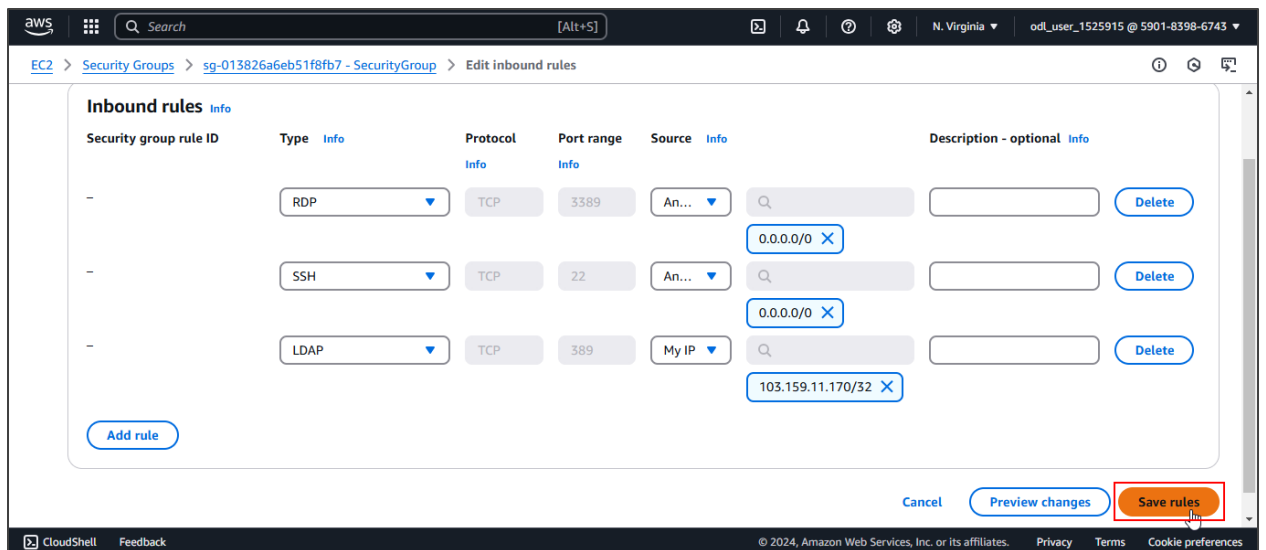
**Tags - optional**  
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.  
No tags associated with the resource.  
[Add new tag](#)  
You can add up to 50 more tags

[Cancel](#) [Create security group](#)

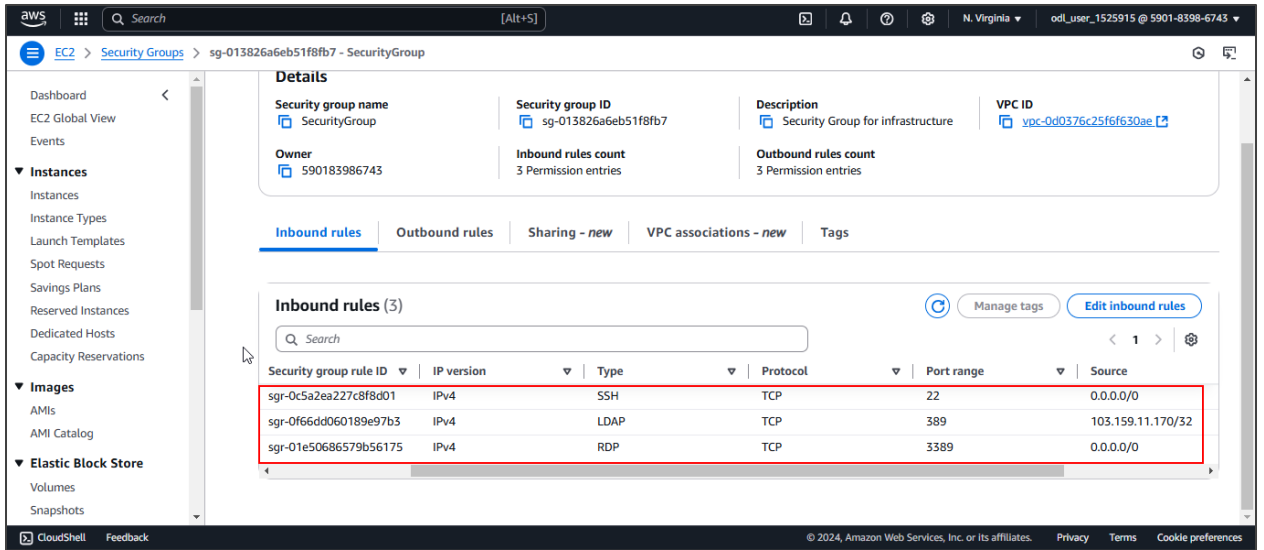
1.8 Click on **Edit inbound rules** and configure inbound rules for **SSH (TCP/22 to 0.0.0.0/0)**, **RDP (TCP/3389 to 0.0.0.0/0)**, and **LDAP (TCP/389 to My IP:<YOUR\_IP>)**



1.9 Click on **Save rules**

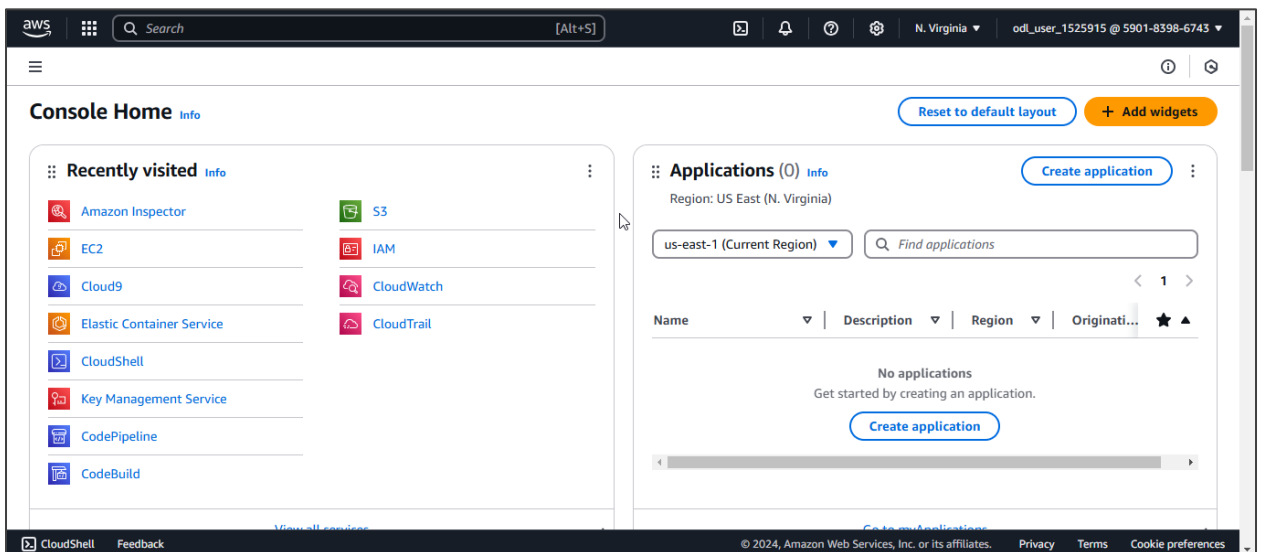


The inbound rules are added successfully.

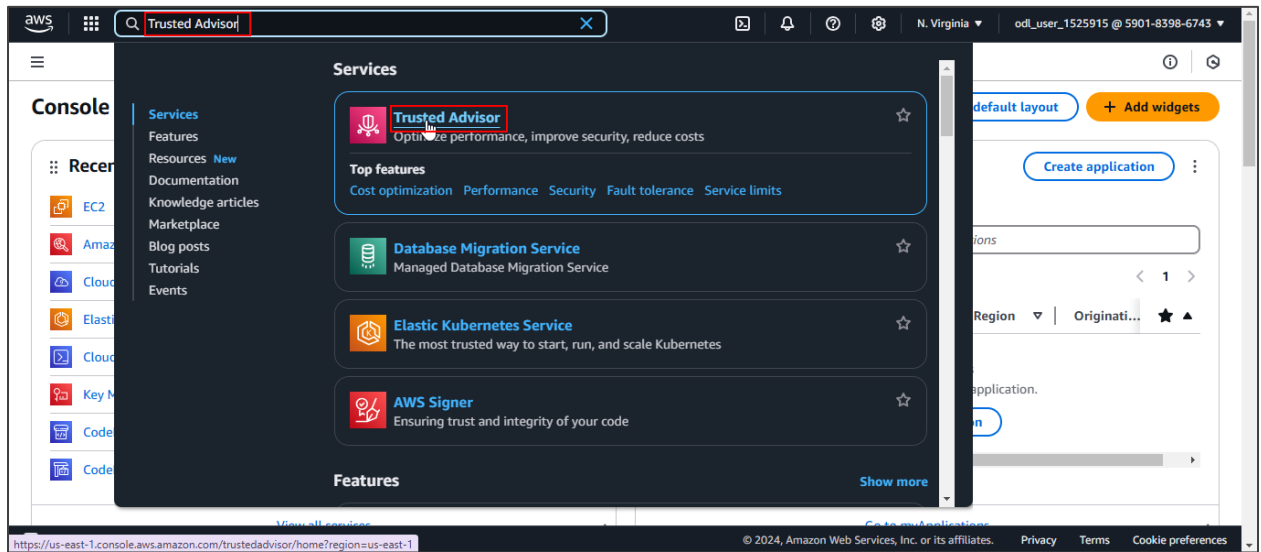


## Step 2: Validate security configurations using AWS Trusted Advisor checks

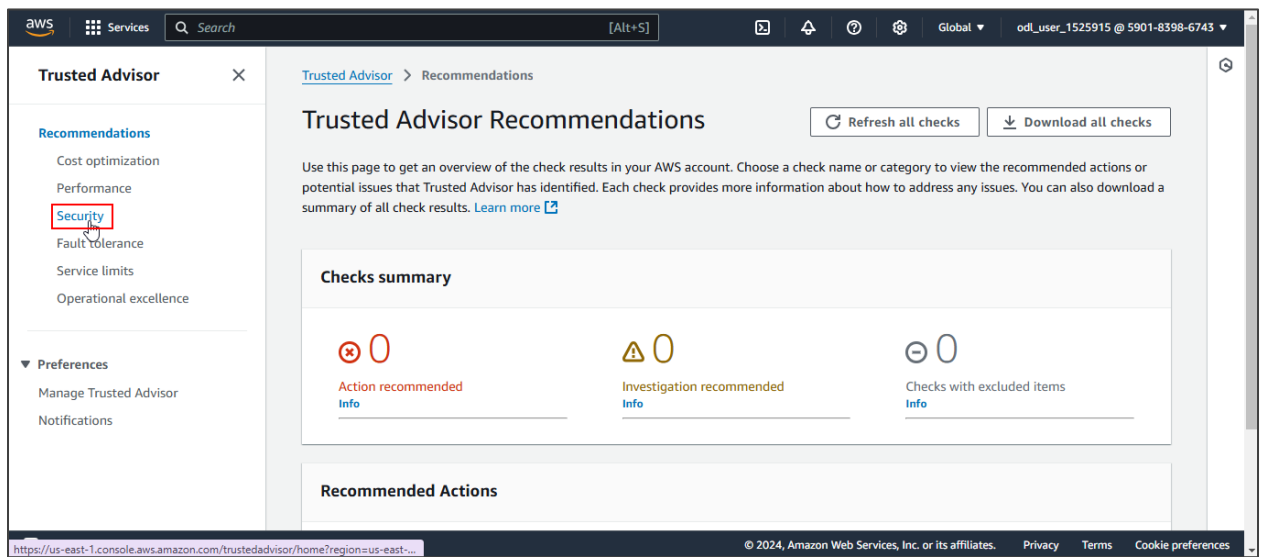
### 2.1 Navigate back to the AWS Console Home page



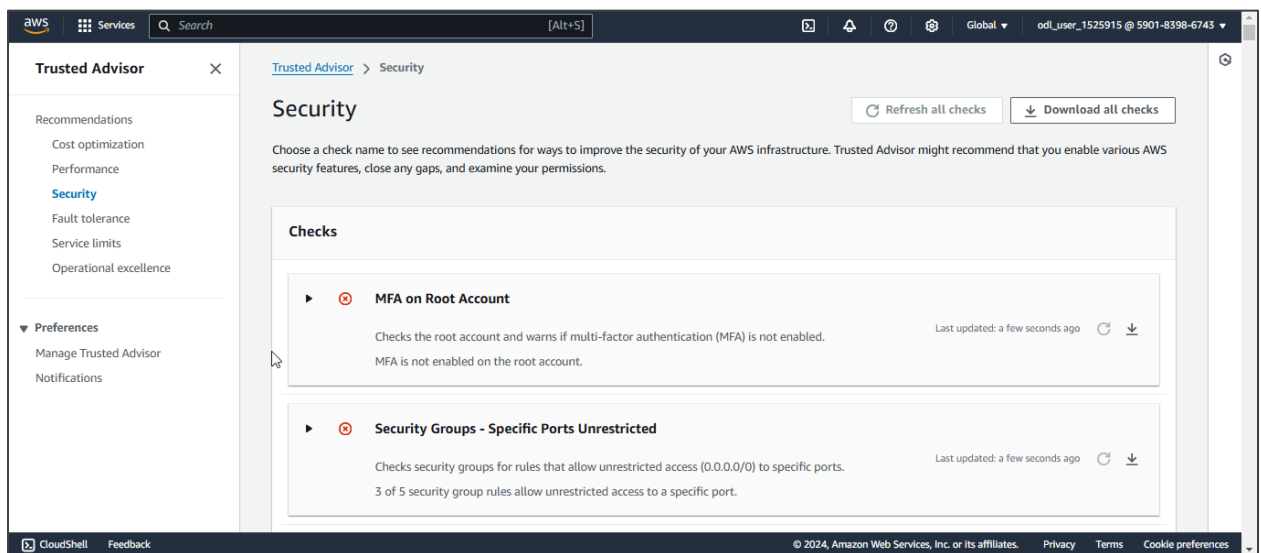
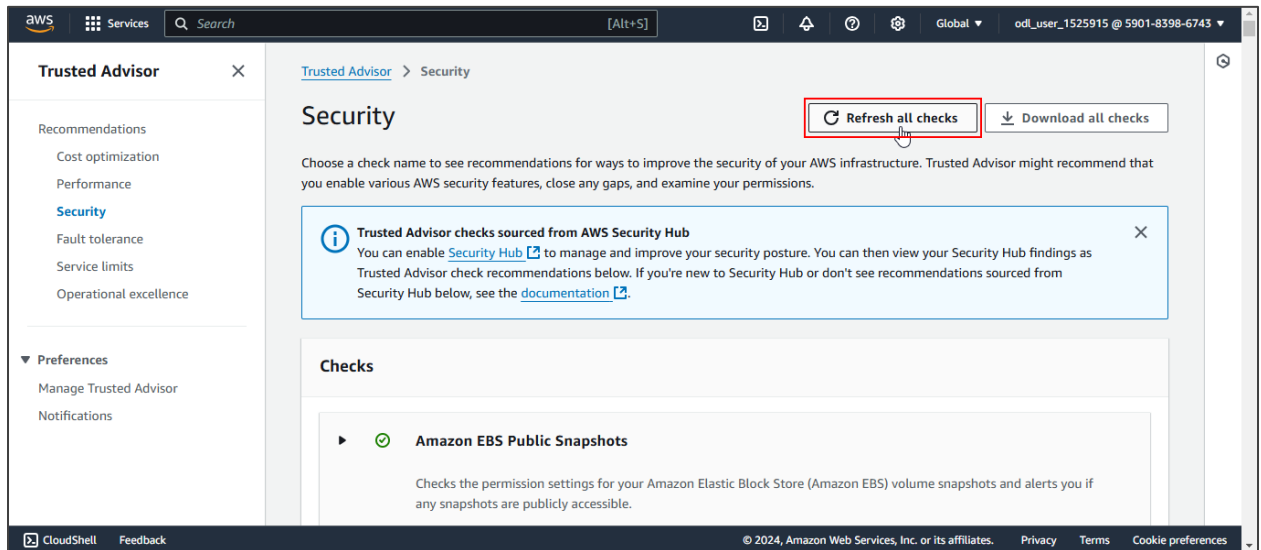
## 2.2 Search for and click on Trusted Advisor



## 2.3 Click on Security

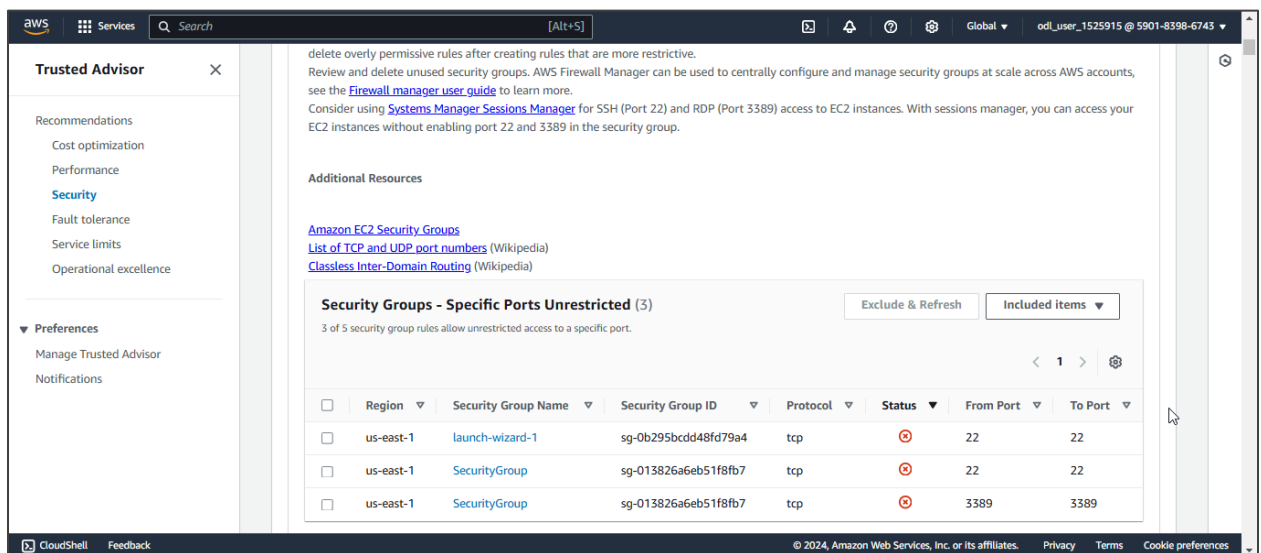
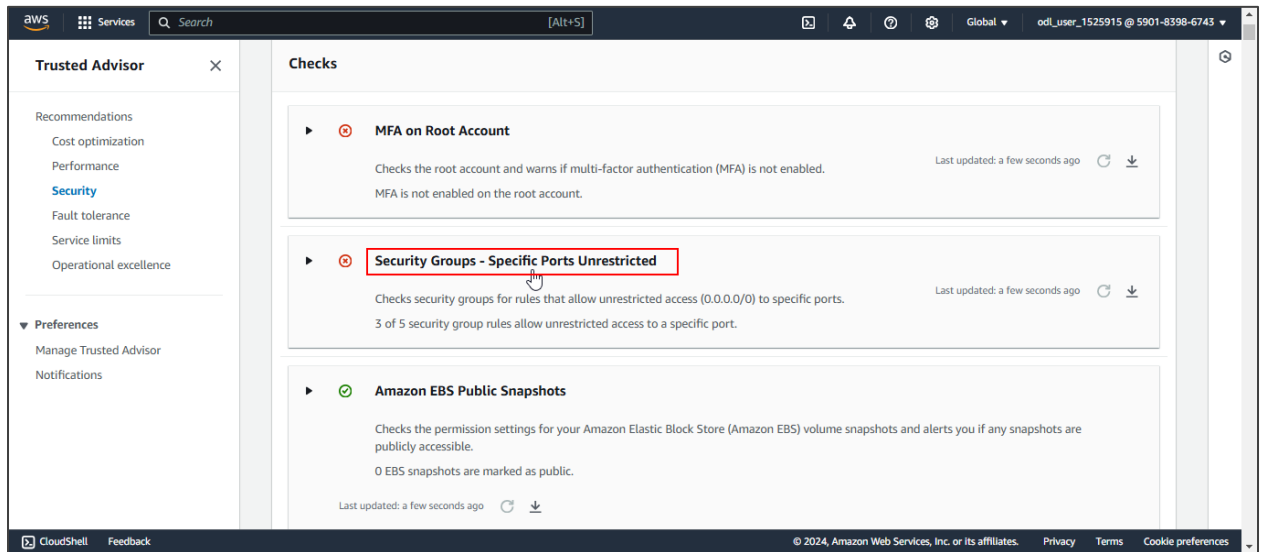


## 2.4 Click on Refresh all checks



**Note:** Some checks may fail after the scan is completed due to non-compliant security configurations.

## 2.5 Click on the desired check to view detailed information



By following these steps, you have successfully created and configured a security group and validated its compliance and configurations using AWS Trusted Advisor checks to ensure secure and optimized AWS resources.